ALPHA · 2026

EUSTRESS.DEV

EUSTRESS ENGINE

What Eustress Can Do for the Center for Innovation

AI-NATIVE · FILESYSTEM-FIRST · BEVY ECS · SOFTWARE-IN-THE-LOOP

ALPHA TESTERS PROGRAM → LAUNCH BRIEFING

// THE WHAT

Eustress is an AI-native engineering orchestrator

for designing, simulating, and shipping physical and digital products.

01

ARCHITECT

Enforces standards. Generates structured specs and telemetry from intent.

02

WATCHMAN

Monitors live state, logs, and SiTL signals via the Model Context Protocol.

03

REPAIRMAN

Ingests Watchman alerts. Diagnoses root cause. Proposes architectural fixes.

UNIVERSE = FOLDER → SPACE = FOLDER → PROTOTYPE = FOLDER → EVERY ARTIFACT IS A FILE

// THE WORKSHOP

The Workshop

Agentic AI chat, embedded in the engine.

MULTI-AGENT

Architect, Watchman, Repairman pillars run as cooperating agents.

TOOL-AWARE

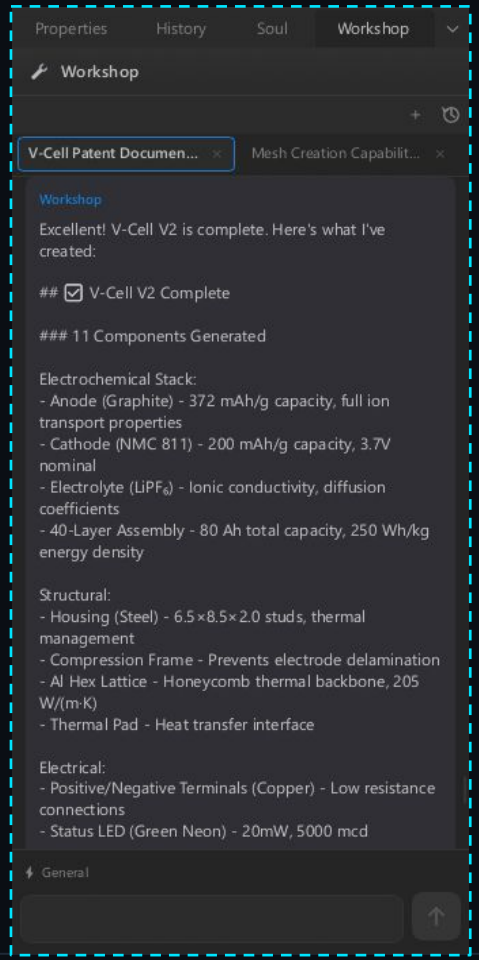
Every chat turn can invoke MCP tools, file operations, or simulations.

FILE-FIRST

Conversations write directly into Universes, Spaces, and Prototypes.

AUDITABLE

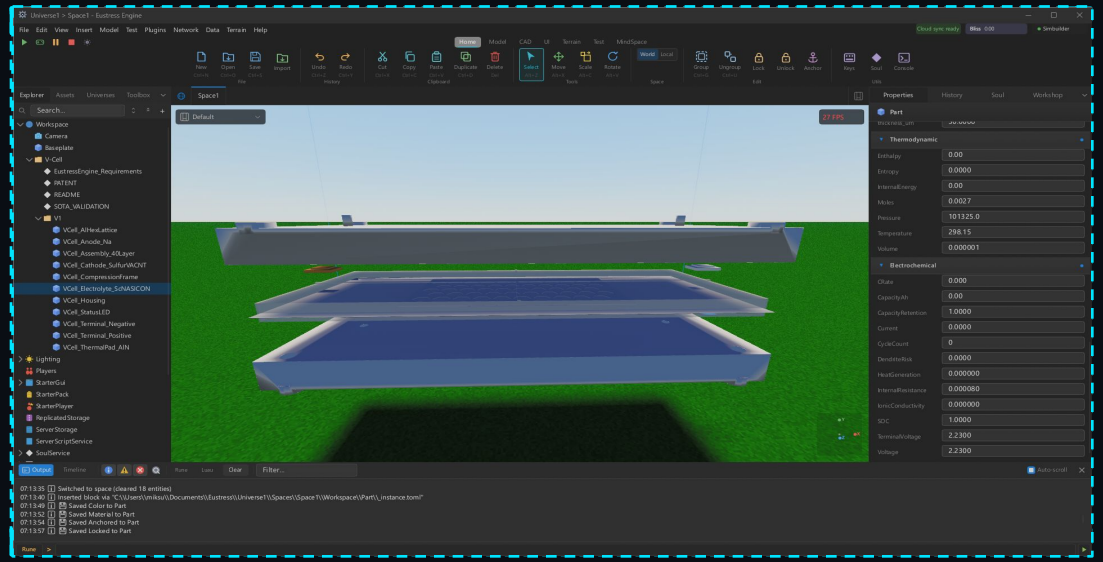
Central log of every tool call, decision, and Type 1 / Type 2 gate.



// THE IDE

Eustress IDE

Universe-aware editor. Bevy ECS hot-reload. Built on filesystem semantics.



UNIVERSE TREE

Browse Universes, Spaces, and Prototype branches as a unified file tree.

ECS INSPECTOR

Live entity/component/system view. Edit components, see Bevy World update in place.

SiTL HARNESS

Run scenarios. Diff prototypes. Promote to HIL only when feedback gates pass.

MCP CONSOLE

Every tool call, every cost, every audit trail — visible and replayable.

// MODEL CONTEXT PROTOCOL

MCP — the bridge between intent and action

Every external capability — file ops, simulations, builds, deploys — speaks Model Context Protocol.

INTENT

Workshop chat
or IDE action

→

MCP

Typed tool call
with full context

→

RESOURCE

Filesystem · ECS
Simulator · Build

// EXAMPLE CAPABILITIES EXPOSED VIA MCP

fs.read	fs.write	ecs.spawn	sim.run	scenario.load
build.cargo	telemetry.tail	mcp.audit	git.branch	feedback.diff

// LANGUAGE SERVER PROTOCOL

LSP — language intelligence everywhere

Eustress speaks LSP so every Universe gets first-class diagnostics, completion, and refactoring.

RUST · BEVY

rust-analyzer integration. Bevy ECS schema-aware completion.

RUNE

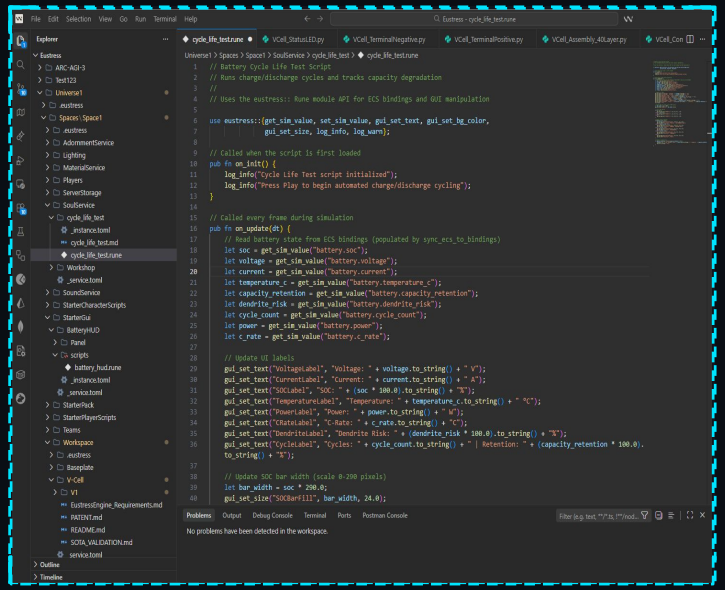
VM Pooled Rust DSL capable of high throughput and scripted actions.

INLINE TELEMETRY

Hover any system to see live SiTL telemetry from the last run.

AGENT-ASSISTED

Repairman proposes LSP code-actions inline when Watchman flags drift.



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PRACTICAL APPLICATIONS

What Eustress Builds Today

// USE CASES · PROTOTYPE LATTICES · SHIPPING PRODUCTS

// VOLTEC · VCELL

Voltec VCell

Battery cell prototype lattice generated and simulated end-to-end.

Targeting ~1000 Wh/kg when the industry standard is 300.

PROTOTYPE LATTICE

v0001	**BASELINE** 21700 form factor, NMC chemistry, BMS reference firmware
v0002	**COST-REDUCED** BOM-trimmed cathode, shared-tooling can geometry
v0003	**ENERGY-DENSE** Silicon-anode blend, revised thermal interface
v0004	**FAST-CHARGE** Tab geometry pass for 4C sustained charge
v0005	**SAFETY-HARDENED** Vent path, separator coating, abuse-test profile

SiTL OUTCOMES

CYCLES SIMULATED	12,400
SCENARIOS PER BRANCH	47
FAILURES CAUGHT PRE-HIL	31
BRANCHES PROMOTED	2 of 5
TURNAROUND	1 hour

// APPLICATION GRID

Other Active Universes

The same lattice works across brands, functions, and industries.

KIRKLAND

/FoodPreparation

Bulk-duty kitchen appliances, pallet-stress simulated

VITAMIX

/BeverageSystems

7-yr duty-cycle blender, member-household scenarios

TRAEGER

/ThermalCooking

Pellet-grill firmware tuning across weather profiles

DYSON

/AirAndClimate

Whole-home IAQ stack with zoned ventilation logic

LG

/ColdStorage

Variable-load fridge compressor across 8,760 hrs/yr

VOLTEC

/Batteries

VCell chemistry & form-factor lattice (case study)

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THE DREAM

What we want Eustress to become

// FRONTIER · AMBITIOUS · ALPHA-TESTER TERRITORY

// VISION 01

DREAM 01

A Personal R&D Lab of Infinite Size

Every fabrication tool, every measurement, every chemistry — addressable from one chat.

FABRICATION

CNC, FDM/SLA print, laser cut, reflow — one job graph.

MEASUREMENT

Scope, logic analyzer, microscope, thermal cam streamed into ECS.

MATERIALS

Chemistry sandbox: simulate before you mix, measure before you keep.

INFINITE

Local rigs + remote cloud fabs federated as one Universe.

// VISION 02

DREAM 02

A Simulation Engine for Civic Decisions

Council-table-grade modeling for the choices that shape neighborhoods.

ZONING

Rezone a parcel — see school enrollment, tax base, and traffic shift in real time.

INFRASTRUCTURE

Bike lane, bus route, stormwater — model first, build once.

BUDGET

Trade departments against projected ten-year outcomes, not next year's politics.

EMERGENCY

Pre-rehearsed responses for fire, flood, outage — drill in software, save in steel.

// VISION 03

DREAM 03

Household Garbage Incinerator + Compactor

Closed-loop combustion, scrubbed exhaust, compaction pre-stage — designed inside Eustress.

01

COMBUSTION

Catalytic afterburner. Flue chemistry simulated across 100k input mixes.

02

SCRUBBING

Particulate + acid-gas capture. Filter life modeled to the household profile.

03

COMPACTION

Pre-stage for non-combustibles. Crusher torque vs. waste-density curve.

04

SAFETY ENVELOPE

Lithium-cell ingress detection. Thermal runaway interlocks. Audit log.

// SiTL FIRST · HIL ONLY ON GREEN · ZERO BAD COMBUSTION REACHES YOUR GARAGE

ALPHA · 2026

Join the Alpha

Build the future of physical and digital products with us.

WHO

Hardware founders, simulation engineers, civic technologists, makers with unusual problems.

WHAT

Early access to the engine, IDE, MCP toolkit, and the Workshop— plus a private channel with the founder via Slack.

HOW

Build a Universe. Ship prototypes. Tell us what's broken. We build with you.

→ EUSTRESS.DEV / ALPHA